Set notation and sample space

1 List all the possible outcomes when a coin is flipped.

2 A standard six-sided die is rolled.

- **a** List the sample space.
- **b** List the sample space for rolling a number strictly less than 3.
- **c** List the sample space for rolling a number divisible by 3.
- **d** List the sample space for rolling an even number.

3 Uther rolled a standard six-sided die.

- **a** List all the numbers that the die may land on.
- **b** Uther rolled a number less than 3. List all the numbers that he could have rolled.



4 List all the elements in the following sets:

- **a** $\{x : x \text{ is a positive even integers that are less than } 16\}$
- **b** $\{x : x \text{ is an integer between } -8 \text{ and } -3 \text{ (not inclusive)}\}$
- **c** $\{x : x \text{ is an odd whole number less than } 13\}$
- **d** $\left\{1, \frac{1}{10}, \frac{1}{100}, \ldots, \frac{1}{100\,000}\right\}$
- **e** $\{2, 4, 8, \ldots, 64\}$

5 Consider the following sets:

$$A = \{1, 2, 4, 8, 16\}, \quad B = \{1, 2, 4, 8\}$$

List all the elements in:

- **a** A or B
- **b** Both A and B

Set notation (Extended)

6 Consider the set $A = \{2, 4, 6, 8\}$. Construct set builder notation for A.

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7 Consider the following sets:

$A = \{8, 12, 16, 20, 24, 28\}, B = \{8, 12, 16, 28\}, C = \{12, 16\}$, and $D = \{4, 8, 12, 16, 20, 24, 28\}$

Determine whether the following statements are true or false:

a $B \subset D$ **b** $C \subset B$ **c** $A \subset B$ **d** $\emptyset \subset D$

8 Write $\subseteq$ or $\nsubseteq$ to make each of the following statements true:

a $\{1, 2, 4\} \square \{1, 2, 4, 7, 8\}$ **b** $\{8, 9\} \square \{3, 4, 7, 8, 9\}$

c $\{3, 4, 9\} \square \{2, 4, 7, 8, 9\}$ **d** $\{1, 3, 8\} \square \{1, 4, 7, 8\}$

9 Write $\in$ or $\notin$ to make each of the following statements true:

a $5 \square \{2, 5, 6, 9\}$ **b** $11 \square \{4, 8, 12, 14\}$

10 Consider the following sets:

- $A = \{\text{letters of the word 'parallelogram'}\}$
- $B = \{\text{letters of the word 'gallop'}\}$
- $C = \{\text{letters of the word 'llama'}\}$

a State the set of elements contained in both A and B.

b Is B a proper subset of A?

c State the set of elements contained in A or C.

11 Explain the meaning of "the complement of set A."

12 A set is defined as $A = \{x : 0 \leq x \leq 5 \text{ and } x \text{ is an integer}\}$.

a List the elements of set A. **b** Is set $B = \{2, 3, 5\}$ a subset of set A?

13 Consider the sets $A = \{x : -6 < x < 6 \text{ and } x \text{ is an integer}\}$ and $B = \{x : -1 \leq x \leq 10 \text{ and } x \text{ is an integer}\}$.

a Find $n(A)$. **b** Find $n(B)$. **c** Find $n(A \cap B)$. **d** Find $n(A \cup B)$.

14 State whether the following points are in the set $A = \{(x, y) : y = 5x - 7\}$:

a $(9, 40)$ **b** $(2, 3)$ **c** $(0, 7)$ **d** $\left(\frac{3}{2}, \frac{1}{2}\right)$

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15 Consider the sets $A = \{(x, y) : y = 3x - 4\}$ and $B = \{(x, y) : y = 2x + 6\}$.

a Explain why $n(A \cap B) = 1$.

b Find $A \cap B$.

Operations on sets

16 Explain the meaning of the following statements:

a Union of set A and set B.

b Intersection of set A and set B.

17 P and Q are sets of vegetable types:

$P = \{\text{carrots, cauliflowers, beans}\}$; $Q = \{\text{cauliflowers, potatoes}\}$

There are no other vegetable types in universal set U.

a Is $P \cup Q$ the set of all vegetable types?

b List the elements in the set $P \cap Q$.

18 Suppose set $A = \{3, 4, 5, 6, 7\}$ and set $B = \{3, 7, 8, 9\}$. Find $A \cap B$.

19 List the elements of $A \cap B$ given the following sets:

a $A = \{\text{even numbers}\}$ and $B = \{\text{square numbers less than } 100\}$.

b $A = \{\text{multiples of } 5\}$ and $B = \{\text{positive numbers less than } 50\}$

20 If A is the set of factors of 24, and B is the set of factors of 36, then list the elements of:

a $B \cup A$

b $A \cap B$

21 Set A is the set of possible outcomes from rolling a standard die, and set B is the set of possible outcomes from rolling an eight-sided die. List the elements of the following sets:

a A

b B

c $A \cap B$

d $A \cup B$

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Complement of sets (Extended)

22 P and Q are sets of flower varieties:

$P = \{\text{roses, lillies, daisies}\}$; $Q = \{\text{lillies, sunflowers}\}$

There are no other flower varieties in universal set U.

a List the elements in the following sets:

i Q' **ii** $Q \cap P$ **iii** $Q \cup P$ **iv** $Q' \cap P$

b Is $(P \cup Q)'$ empty?

23 The sets $V = \{21, 8, 30, 9, 28\}$ and $W = \{21, 8, 30, 9, 28, 7, 13, 12, 26\}$ are such that there are no other elements outside of these two sets.

a Is V a proper subset of W?

b State $n(V)$, the order of V.

c List the elements of V'.

d List the elements of the universal set.

e Find W'.

24 Consider the following sets:

$$A = \{1, 2, 3, 4, 5, 6, 7\}, B = \{1, 2, 3, 4\}$$

If there are no elements contained outside of these sets, find:

a B'

b A'

25 Consider the following sets:

$A = \{\text{people who like football}\}$

$B = \{\text{people who like softball}\}$

$C = \{\text{people who like swimming}\}$

$D = \{\text{people who do not like any of these}\}$

a Describe set B'.

b Describe set D'.

26 Consider the sets $A = \{x : -7 \leq x \leq 10 \text{ and } x \text{ is an integer}\}$, $B = \{-2, 4, 8\}$ and $C = \{3, -5, 11\}$.

a Is set B a subset of set A?

b Is set C a subset of set A?

c List the elements in the set $A \cap C$.

d List the elements in the set $A' \cap C$.

e Find $n(A \cup C)$.

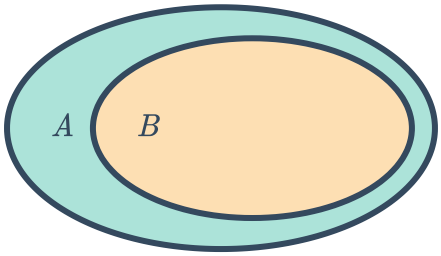
f Find $n(A \cap C')$.

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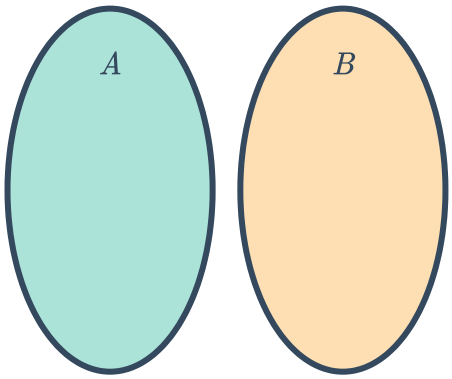
Venn diagrams

27 For each of the following, identify the Venn diagram that best represents the sets A and B:

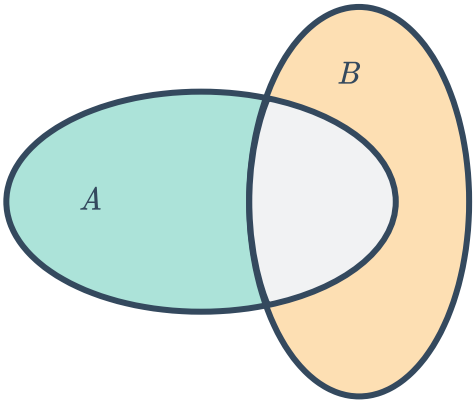
Venn diagram 1



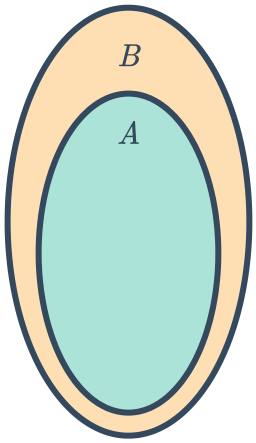
Venn diagram 2



Venn diagram 3



Venn diagram 4



a $A = \{\text{Earth}\}$, $B = \{\text{Planets}\}$

b $A = \{6, 8, 4, 2, 7\}$, $B = \{3, 6, 1\}$

c $A = \{\text{w}, \text{i}, \text{n}, \text{d}\}$, $B = \{\text{e}, \text{a}, \text{r}, \text{t}, \text{h}\}$

d $A = \{\text{Animals found in Australia}\}$, $B = \{\text{Animals found in NSW}\}$

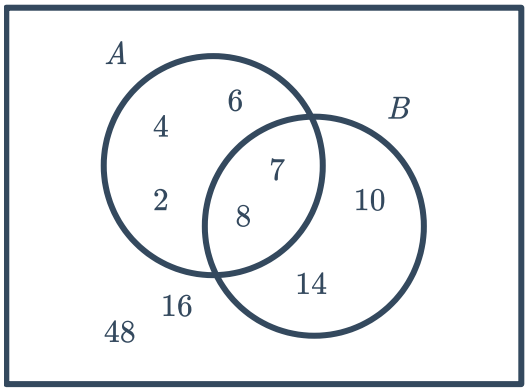
28 We are interested in the colour of a card randomly drawn from a standard deck. Draw a Venn diagram to illustrates this.

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29 Consider the Venn diagram:

List the elements in:

a $A \cap B$ **b** $A \cup B$

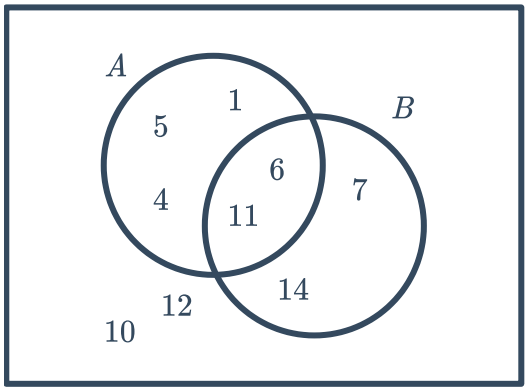


Venn diagrams (Extended)

30 Consider the Venn diagram:

List the elements in:

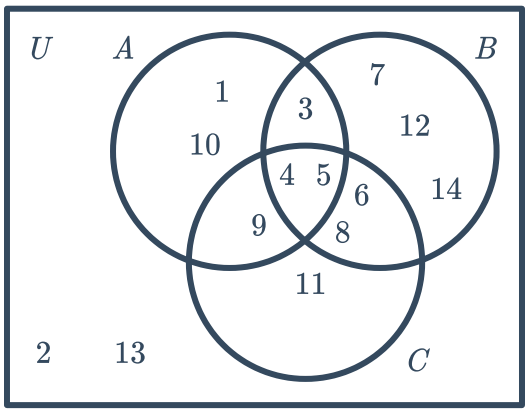
a A' **b** B'



31 Consider the Venn diagram:

List the elements in:

a A **b** U **c** B'

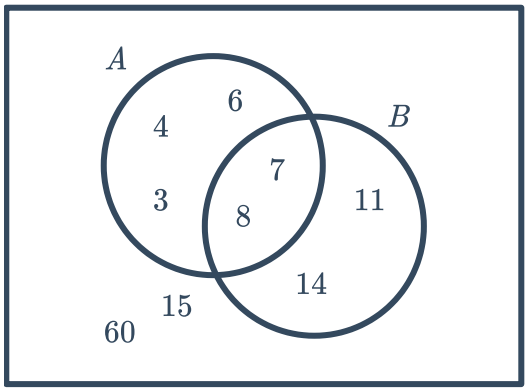


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32 Consider the Venn diagram:

List the elements in:

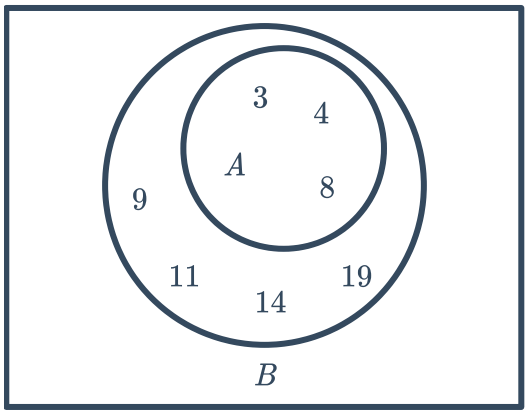
a $A \cap B'$ **b** $(A \cup B)'$



33 Consider the following Venn diagram:

a Is A a subset of B?

b Is A a proper subset of B?



34 Consider the following diagram:

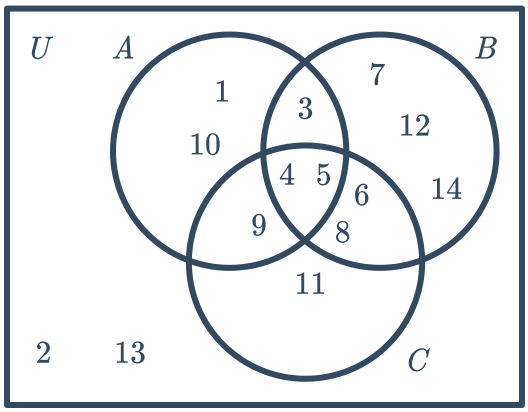
a Find the set $A \cap C$.

b Find the set $(B \cap C)'$.

c Find the set $A \cap B \cap C$.

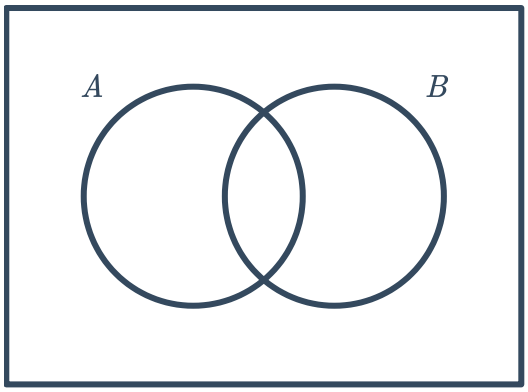
d Find the set $A \cap (B \cup C)$.

e Find the set $(A \cap B)'$.



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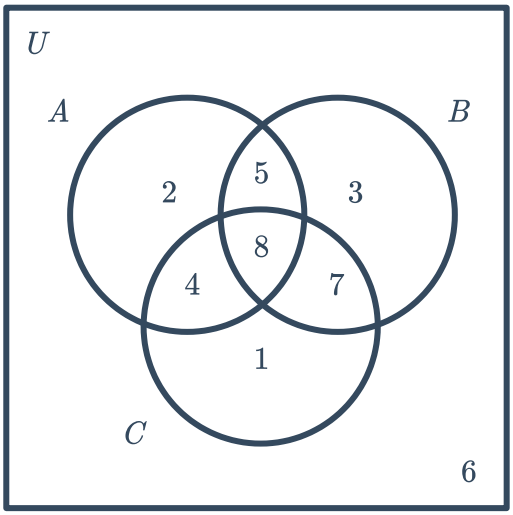
35 Consider the following Venn diagram:



a Is $(A \cap B)'$ equal to $A' \cup B'$ for all sets?

b Is $A' \cap B'$ equal to $A \cup B'$ for all sets?

36 Consider the following Venn diagram:



Find the elements in the following:

a $A \cap B' \cap C'$

b $A \cap B \cap C'$

c $A' \cap B \cap C'$

d $A \cap B' \cap C$

e $A \cap B \cap C$

f $A' \cap B \cap C$

g $A' \cap B' \cap C$

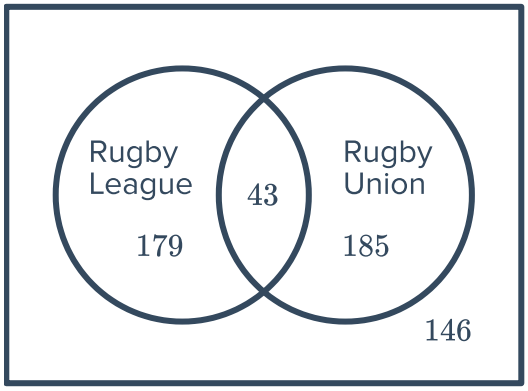
h $A' \cap B' \cap C'$

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37 The Venn diagram shows the number of students in a school playing Rugby League (L), Rugby Union (U), both or neither.

a Complete the table of values below:

	Play Rugby League	Don't play Rugby League
Play Rugby Union		
Don't play Rugby Union		



b Find the following:

i $n(L \cap U')$ **ii** $n(L)$

iii $n(U)$ **iv** $n(U \cap L')$

v $n(L')$ **vi** $n(U')$

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